



Engineering Electromagnetics

工程电磁场

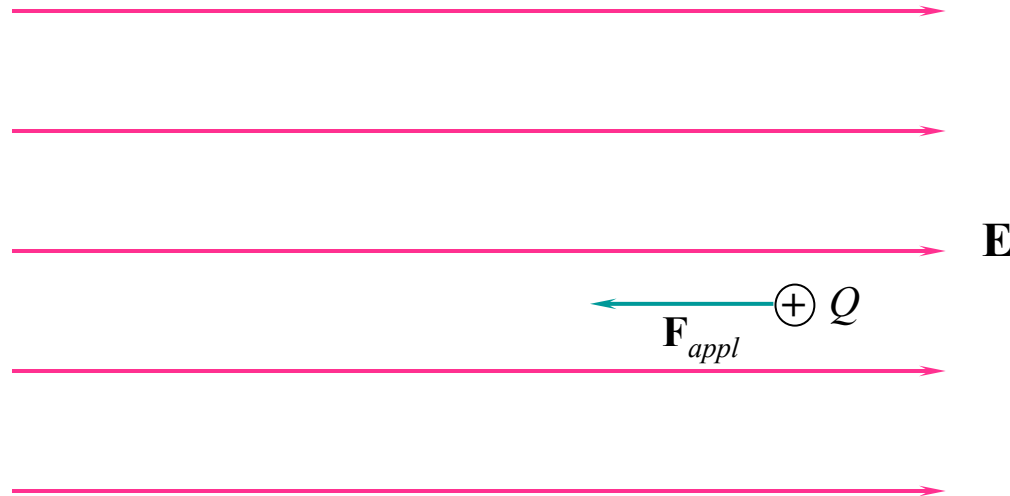
Youwei Jia 嘉有为

8: Energy and Potential 能量与电位



- Point charge in an external field 外场中的点电荷

To move charge Q **against** 逆着 the electric field, a **force** must be applied that **counteracts** 克服 the **force** on Q that arises from the field:



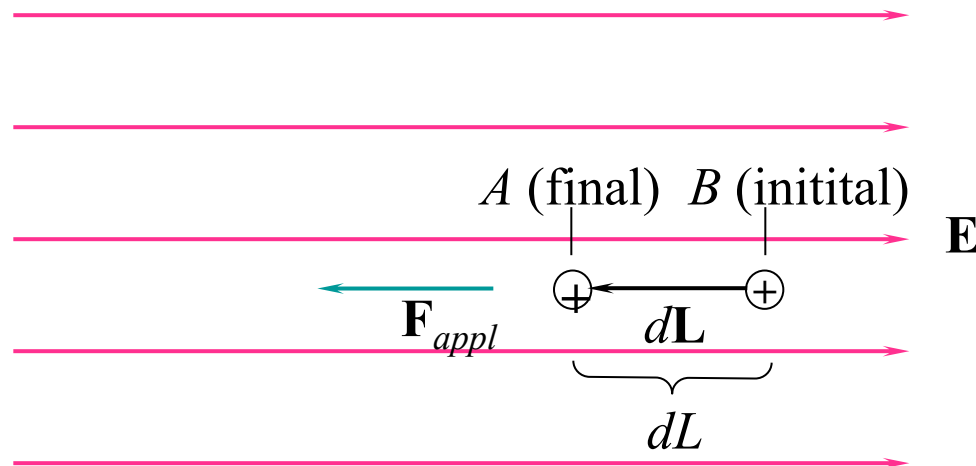
$$\mathbf{F}_{appl} = -Q \mathbf{E}$$

8: Energy and Potential 能量与电位



- **Differential work 功** done on moving a point charge against an external field

In moving point charge Q from **initial position B** over a **differential distance dL** (to **final position A**), the work expended is:



$$dW = F_{appl} dL = QE dL = \underline{-QE \cdot dL} \text{ [J]}$$

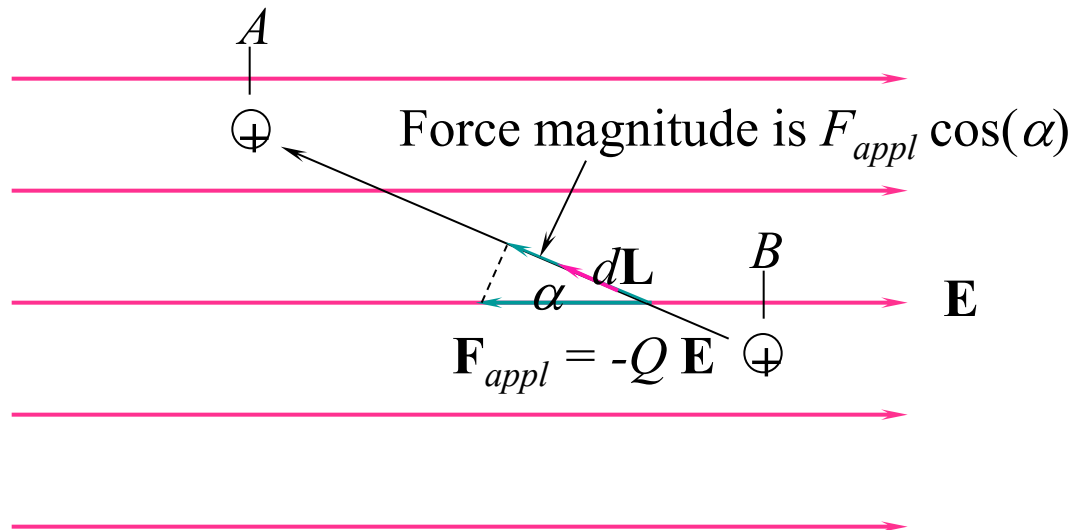
Note: Herein, the path is along an electric field line (in the opposite direction), and over the differential path length, the field can be assumed constant.

8: Energy and Potential 能量与电位



- Forcing a charge against the field in **an arbitrary direction 任意方向**

What matters now is **the component of force in the direction of motion.**



Differential work in moving charge Q through distance dL will be:

$$dW = F_{appl} \cos(\alpha) dL = -QE \cdot d\mathbf{L} \text{ [J]}$$

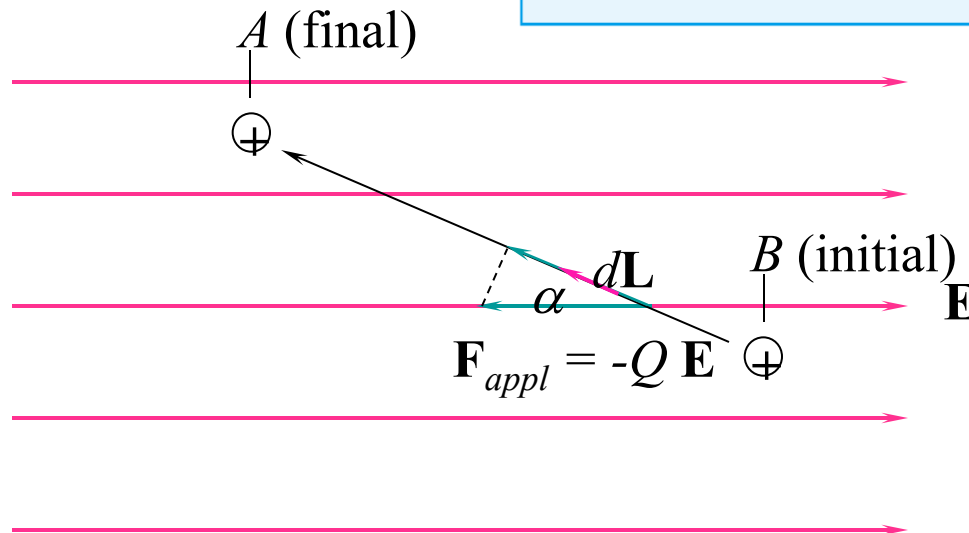
8: Energy and Potential 能量与电位



- Forcing a charge against the field in **an arbitrary direction 任意方向**

Total work done--All differential work contributions along the path are summed to give:

$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$



Note: The path must be specified before the integral can be evaluated.

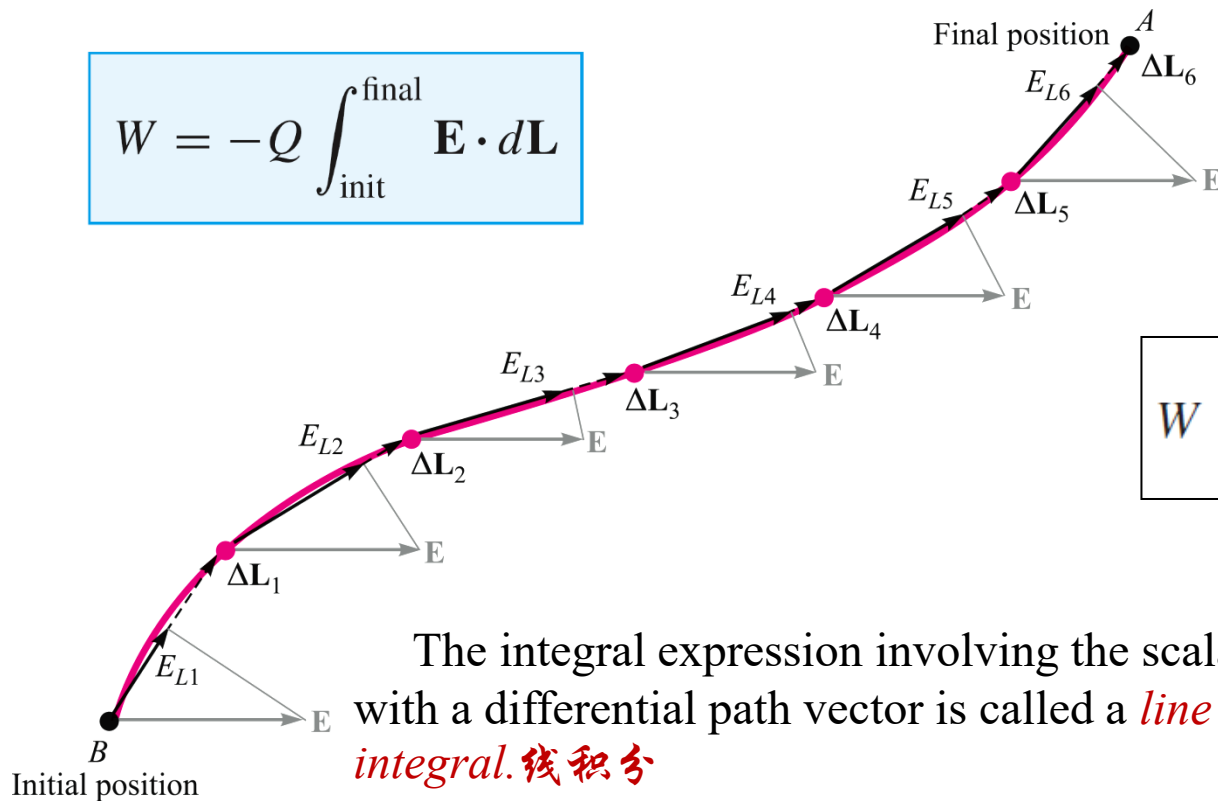
8: Energy and Potential 能量与电位



- Total work done over an arbitrary path

The integral expression for work is **completely general**: Any shape path may be taken, with the component of force evaluated on each differential path segment.

$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$



$$W = -Q \int_{\text{init}}^{\text{final}} E_L dL$$

The integral expression involving the scalar product of the field with a differential path vector is called a **line integral** or a **contour integral**. 线积分

8: Energy and Potential 能量与电位

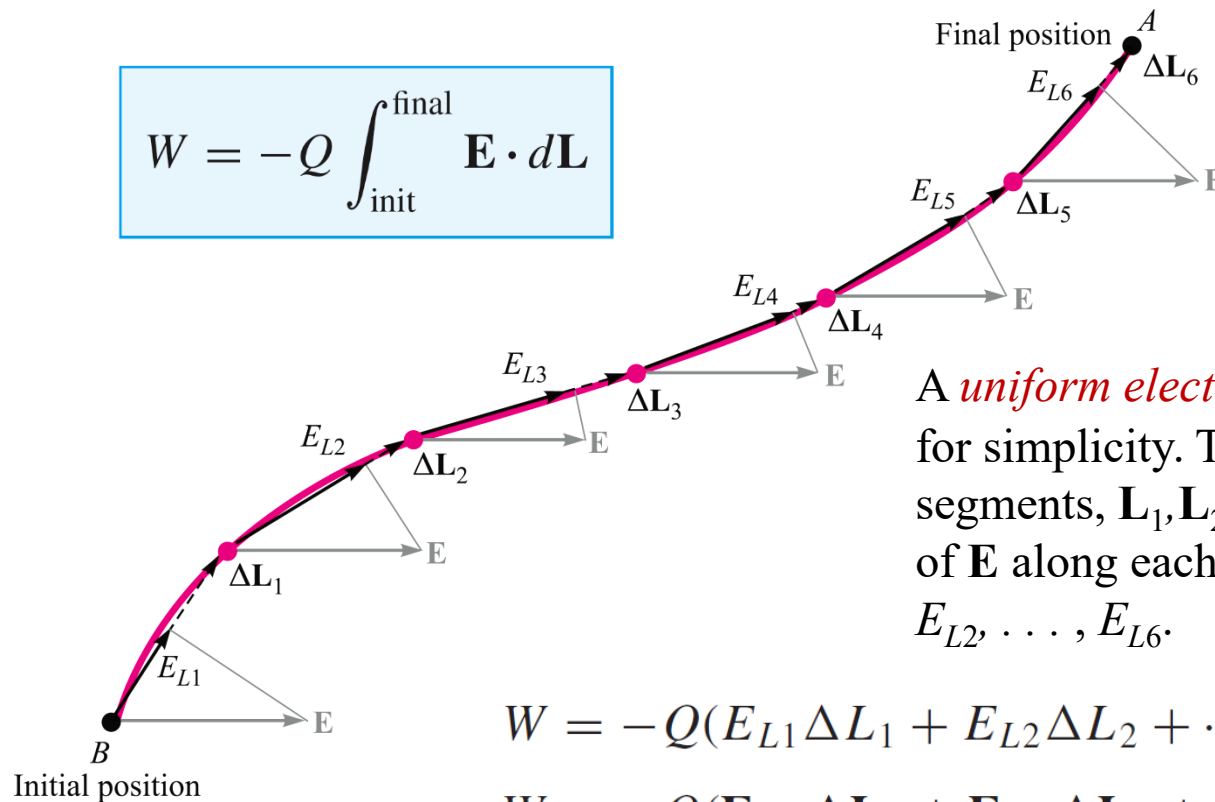


- Total work done over an arbitrary path

The integral expression for work is completely general: Any shape path may be taken, with the component of force evaluated on each differential path segment.

$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

$$W = -Q \int_{\text{init}}^{\text{final}} E_L dL$$



A *uniform electric field* 均匀电场 is selected for simplicity. The path is divided into six segments, $\mathbf{L}_1, \mathbf{L}_2, \dots, \mathbf{L}_6$, and the components of $\mathbf{E}$ along each segment are denoted by $E_{L1}, E_{L2}, \dots, E_{L6}$.

$$W = -Q(E_{L1}\Delta L_1 + E_{L2}\Delta L_2 + \dots + E_{L6}\Delta L_6)$$

$$W = -Q(\mathbf{E}_1 \cdot \Delta\mathbf{L}_1 + \mathbf{E}_2 \cdot \Delta\mathbf{L}_2 + \dots + \mathbf{E}_6 \cdot \Delta\mathbf{L}_6)$$

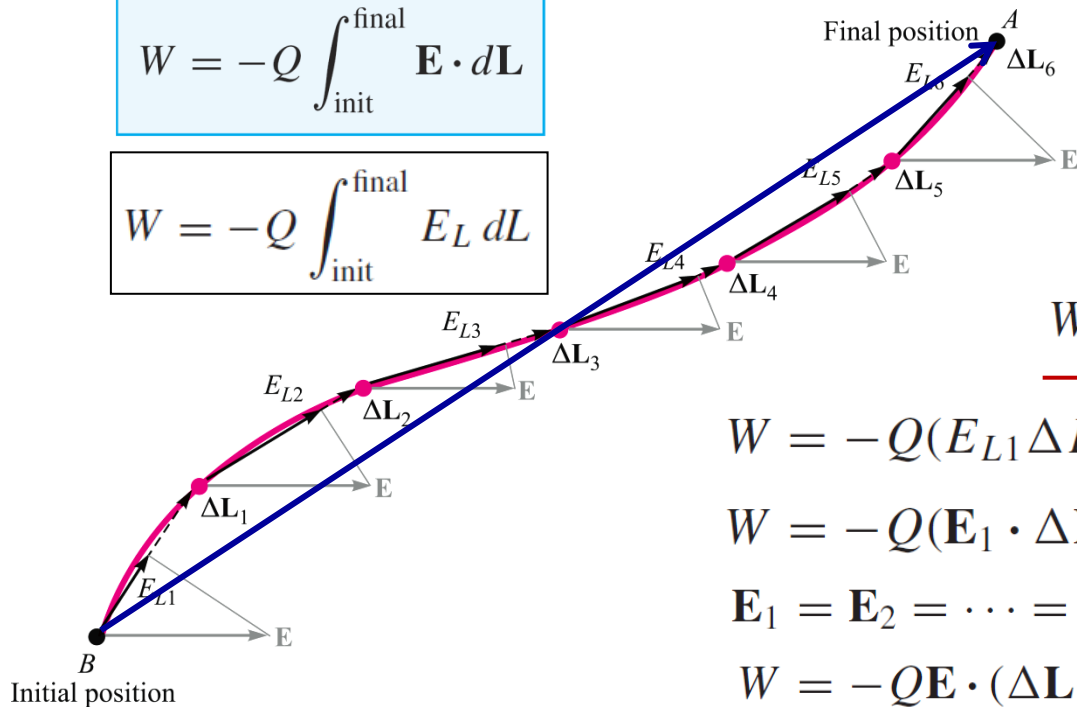
8: Energy and Potential 能量与电位



- Total work done over an arbitrary path

$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

$$W = -Q \int_{\text{init}}^{\text{final}} E_L dL$$



$$W = -Q \int_B^A \mathbf{E} \cdot d\mathbf{L}$$

$$W = -QE \cdot \int_B^A dL$$

$$W = -QE \cdot \mathbf{L}_{BA} \quad (\text{uniform } \mathbf{E})$$

$$W = -Q(E_{L1}\Delta L_1 + E_{L2}\Delta L_2 + \cdots + E_{L6}\Delta L_6)$$

$$W = -Q(\mathbf{E}_1 \cdot \Delta \mathbf{L}_1 + \mathbf{E}_2 \cdot \Delta \mathbf{L}_2 + \cdots + \mathbf{E}_6 \cdot \Delta \mathbf{L}_6)$$

$$\mathbf{E}_1 = \mathbf{E}_2 = \cdots = \mathbf{E}_6$$

$$W = -QE \cdot (\Delta \mathbf{L}_1 + \Delta \mathbf{L}_2 + \cdots + \Delta \mathbf{L}_6)$$

$$W = -QE \cdot \mathbf{L}_{BA} \quad (\text{uniform } \mathbf{E})$$

Note: In a uniform electric field, the work involved in moving a charge does not depend on the particular path we have selected along which to carry the charge.

8: Energy and Potential 能量与电位



- Line Integral Evaluation 线积分计算

We wish to find: $\int_B^A \mathbf{E} \cdot d\mathbf{L}$

where $\mathbf{E} = E_x \mathbf{a}_x + E_y \mathbf{a}_y + E_z \mathbf{a}_z$

and $d\mathbf{L} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$

using these:

$$\int_B^A \mathbf{E} \cdot d\mathbf{L} = \int_B^A (E_x \mathbf{a}_x + E_y \mathbf{a}_y + E_z \mathbf{a}_z) \cdot (dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z)$$

$$= \int_{x_B}^{x_A} E_x dx + \int_{y_B}^{y_A} E_y dy + \int_{z_B}^{z_A} E_z dz$$

8: Energy and Potential 能量与电位



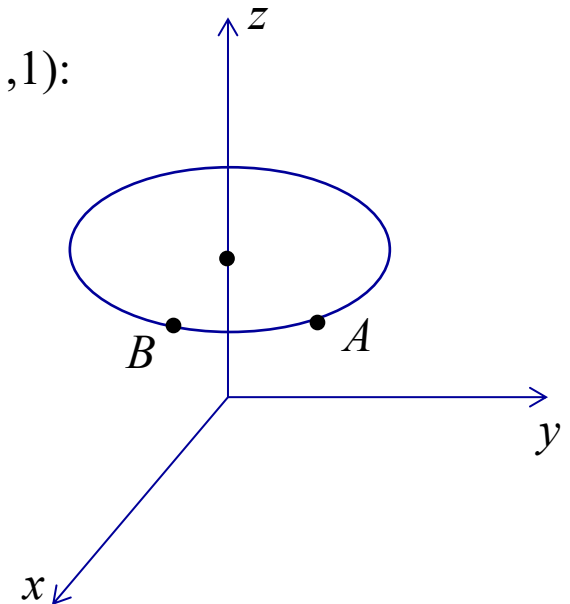
- Example

An electric field is given as: $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$

We wish to find the work done in moving a point charge of magnitude $Q = 2\text{C}$ over the shorter arc of the circle given by $x^2 + y^2 = 1 \quad z = 1$

The initial point is $B(1, 0, 1)$ and the final point is $A(0.8, 0.6, 1)$:

$$\begin{aligned} W &= -Q \int_B^A \mathbf{E} \cdot d\mathbf{L} \\ &= -2 \int_B^A (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z) \cdot (dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z) \\ &= -2 \int_1^{0.8} y dx - 2 \int_0^{0.6} x dy - 4 \int_1^1 dz \end{aligned}$$



This is the basic setup, in which the path has not yet been specified.

8: Energy and Potential 能量与电位



- Example (Cont.'d)

We now have
$$W = -2 \int_1^{0.8} y dx - 2 \int_0^{0.6} x dy - 4 \int_1^1 dz$$

Note that the third integral vanishes because there is no motion along the z direction, and we need to include the y dependence on x in the first integral, and the x dependence on y in the second integral:

Using the given equation for the circular path, $x^2 + y^2 = 1$ $z = 1$, we rewrite the integrals:

$$\begin{aligned} W &= -2 \int_1^{0.8} \sqrt{1-x^2} dx - 2 \int_0^{0.6} \sqrt{1-y^2} dy - 0 \\ &= -\left[x\sqrt{1-x^2} + \sin^{-1} x \right]_1^{0.8} - \left[y\sqrt{1-y^2} + \sin^{-1} y \right]_0^{0.6} \\ &= -(0.48 + 0.927 - 0 - 1.571) - (0.48 + 0.644 - 0 - 0) \\ &= \underline{-0.96 \text{ J}} \end{aligned}$$

8: Energy and Potential 能量与电位

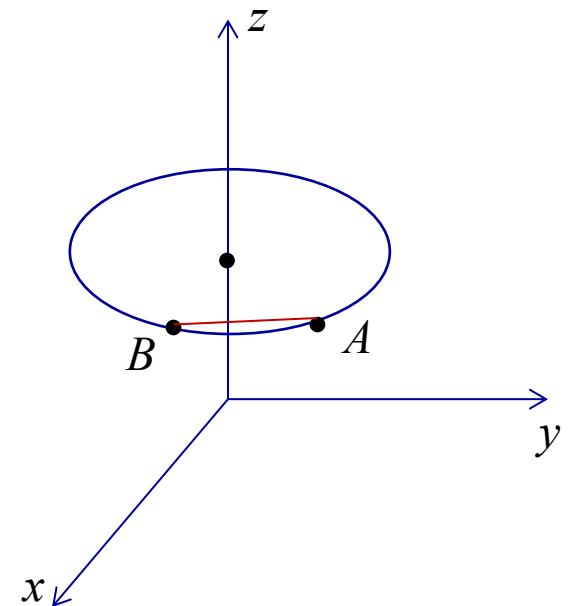


- Example (Cont.'d)

Again find the work required to carry $2C$ from B to A in the same field, but this time use the straight-line path from B to A .

We start by determining the equations of the straight line.
Any two of the following three equations for planes passing through the line are sufficient to define the line:

$$\left\{ \begin{array}{l} y - y_B = \frac{y_A - y_B}{x_A - x_B}(x - x_B) \Rightarrow y = -3(x - 1) \\ z - z_B = \frac{z_A - z_B}{y_A - y_B}(y - y_B) \Rightarrow z = 1 \\ x - x_B = \frac{x_A - x_B}{z_A - z_B}(z - z_B) \end{array} \right.$$



8: Energy and Potential 能量与电位

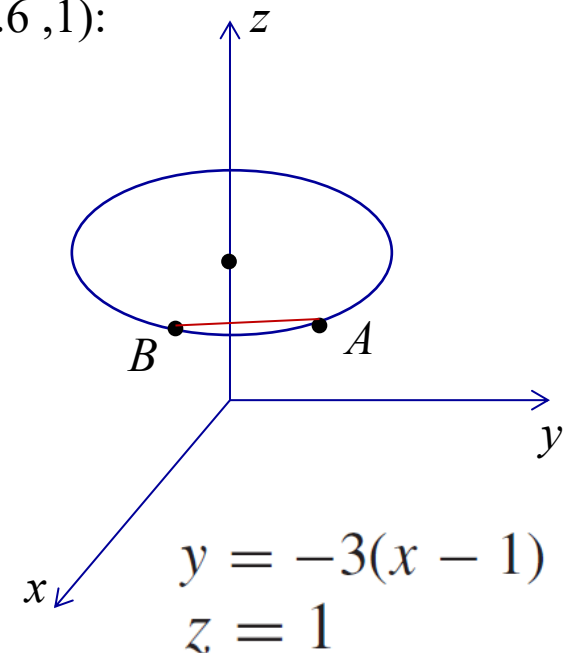


- Example (Cont.'d)

Again find the work required to carry $2C$ from B to A in the same field, but this time use the straight-line path from B to A .

The initial point is $B(1, 0, 1)$ and the final point is $A(0.8, 0.6, 1)$:

$$\begin{aligned} W &= -Q \int_B^A \mathbf{E} \cdot d\mathbf{L} \\ &= -2 \int_B^A (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z) \cdot (dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z) \\ &= -2 \int_1^{0.8} y dx - 2 \int_0^{0.6} x dy - 4 \int_1^1 dz \\ &= 6 \int_1^{0.8} (x - 1) dx - 2 \int_0^{0.6} \left(1 - \frac{y}{3}\right) dy \\ &= \underline{-0.96 \text{ J}} \end{aligned}$$



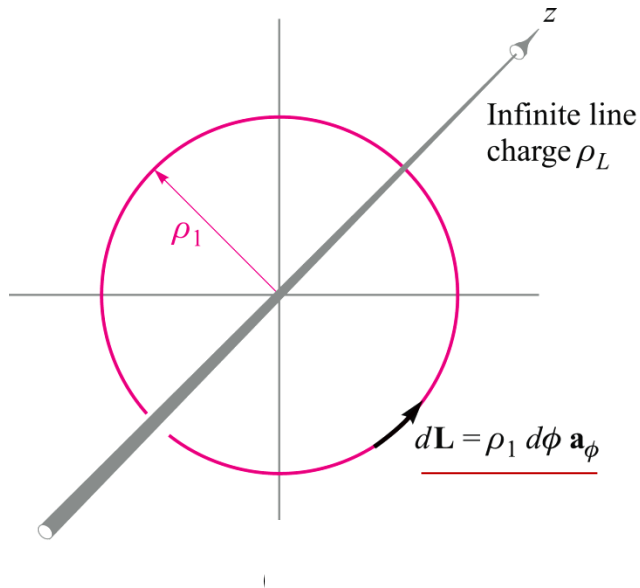
Note: It again demonstrates the statement (unproved) that the work done is independent of the path taken in any electrostatic field.

8: Energy and Potential 能量与电位



- Another Example: Evaluating work within a line charge field

In this example, the work in moving charge Q in a circular path around a line charge is found:



$$\text{where } \mathbf{E} = E_\rho \mathbf{a}_\rho = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho$$

$$\begin{aligned} W &= -Q \int_{\text{init}}^{\text{final}} \frac{\rho_L}{2\pi\epsilon_0\rho_1} \mathbf{a}_\rho \cdot \rho_1 d\phi \mathbf{a}_\phi \\ &= -Q \int_0^{2\pi} \frac{\rho_L}{2\pi\epsilon_0} d\phi \mathbf{a}_\rho \cdot \mathbf{a}_\phi = 0 \end{aligned}$$

8: Energy and Potential 能量与电位



- Another Example: Evaluating work within a line charge field

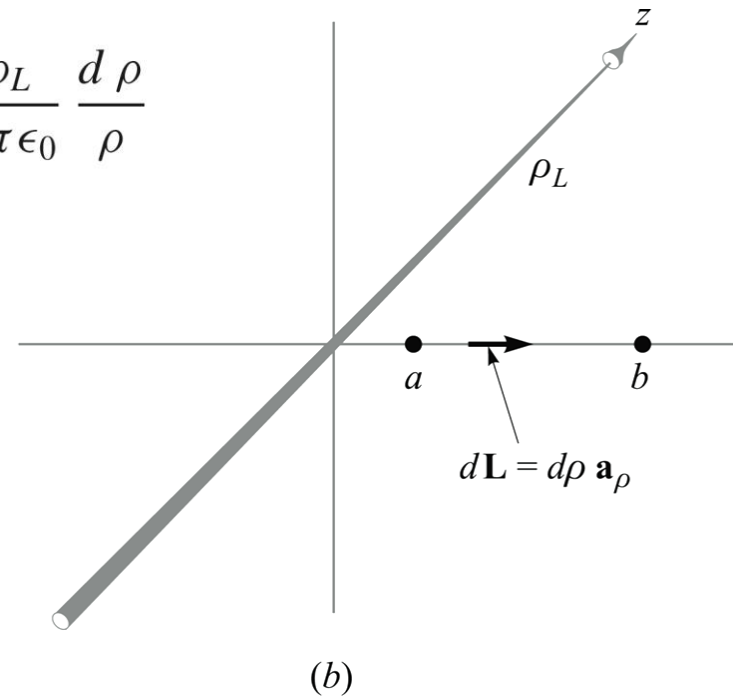
In this example, the work in moving charge Q in radial direction near a Line charge is found:

$$W = -Q \int_{\text{init}}^{\text{final}} \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho \cdot d\rho \mathbf{a}_\rho = -Q \int_a^b \frac{\rho_L}{2\pi\epsilon_0} \frac{d\rho}{\rho}$$

so that finally:
$$W = -\frac{Q\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$

Suppose we carry Q from b to a , We still have $d\mathbf{L} = d\rho \mathbf{a}_\rho$ and show the different direction by recognizing $\rho = b$ as the initial point and $\rho = a$ as the final point,

$$W = -Q \int_b^a \frac{\rho_L}{2\pi\epsilon_0} \frac{d\rho}{\rho} = \frac{Q\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$



8: Energy and Potential 能量与电位



- Differential path lengths in three coordinate systems

$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

$$d\mathbf{L} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z \quad (\text{rectangular})$$

$$d\mathbf{L} = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z \quad (\text{cylindrical})$$

$$d\mathbf{L} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi \quad (\text{spherical})$$

8: Energy and Potential 能量与电位



D4.1. Given the electric field $\mathbf{E} = \frac{1}{z^2}(8xyz\mathbf{a}_x + 4x^2z\mathbf{a}_y - 4x^2y\mathbf{a}_z)$ V/m, find the differential amount of work done in moving a 6-nC charge a distance of $2\text{ }\mu\text{m}$, starting at $P(2, -2, 3)$ and proceeding in the direction $\mathbf{a}_L = (a) -\frac{6}{7}\mathbf{a}_x + \frac{3}{7}\mathbf{a}_y + \frac{2}{7}\mathbf{a}_z$; (b) $\frac{6}{7}\mathbf{a}_x - \frac{3}{7}\mathbf{a}_y - \frac{2}{7}\mathbf{a}_z$; (c) $\frac{3}{7}\mathbf{a}_x + \frac{6}{7}\mathbf{a}_y$.

Ans. -149.3 fJ ; 149.3 fJ ; 0

$$dW = F_{\text{appl}} dL = QE dL = -Q\mathbf{E} \cdot d\mathbf{L} \text{ [J]}$$

8: Energy and Potential 能量与电位



• Definition of Potential Difference

We now have the work done in moving charge Q from initial to final positions. This is the **potential energy 势能** gained by the charge **as a result of this position change**.

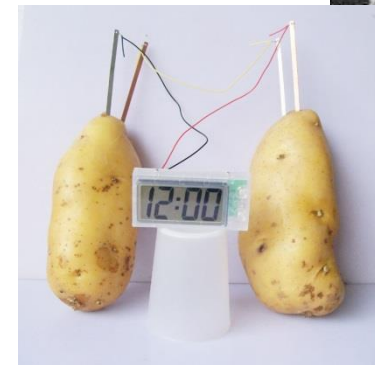
$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

The **potential difference 电位差** is defined as the work done (or potential energy gained) *per unit positive charge*. **外源把单位正电荷从电场中一点移到另一点所做的功**

We express this quantity in units of **Joules/Coulomb 焦耳/库**, or **volts 伏特**:

$$\text{Potential Difference} = \frac{W}{Q} = - \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L} \quad \text{Volts}$$

$$\text{Finally: } V_{AB} = V_A - V_B = - \int_B^A \mathbf{E} \cdot d\mathbf{L}$$



Note: Please compare the height difference in gravitational field with the potential difference in electrostatic field.

8: Energy and Potential 能量与电位



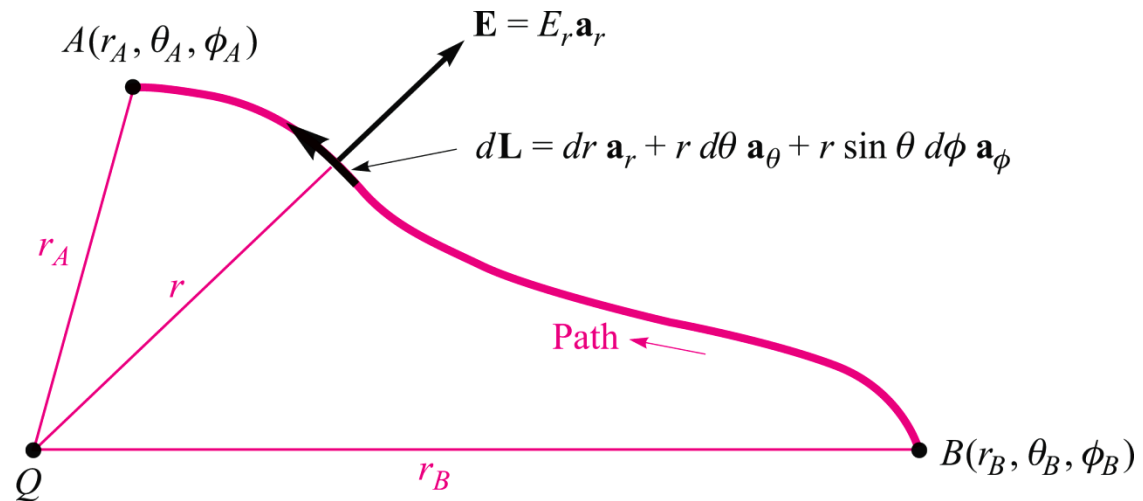
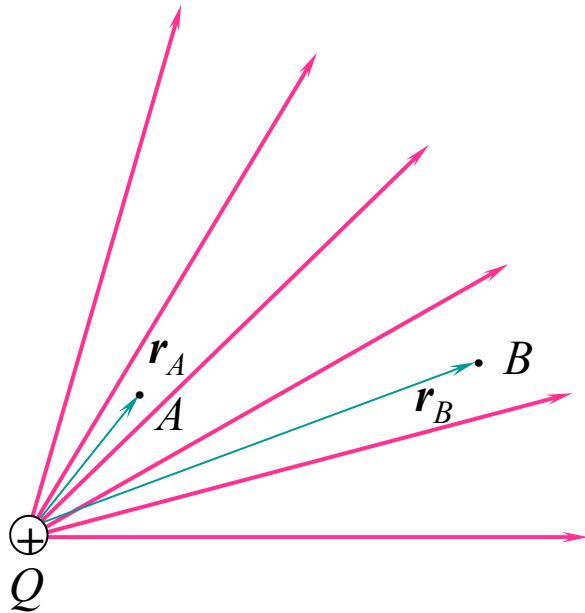
- Potential difference in a point charge field

In this exercise, we evaluate the work done in moving a unit positive charge from point B to point A , within the field associated with point charge Q

where $\mathbf{E} = E_r \mathbf{a}_r = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r$

and where in general:

$$d\mathbf{L} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$



8: Energy and Potential 能量与电位



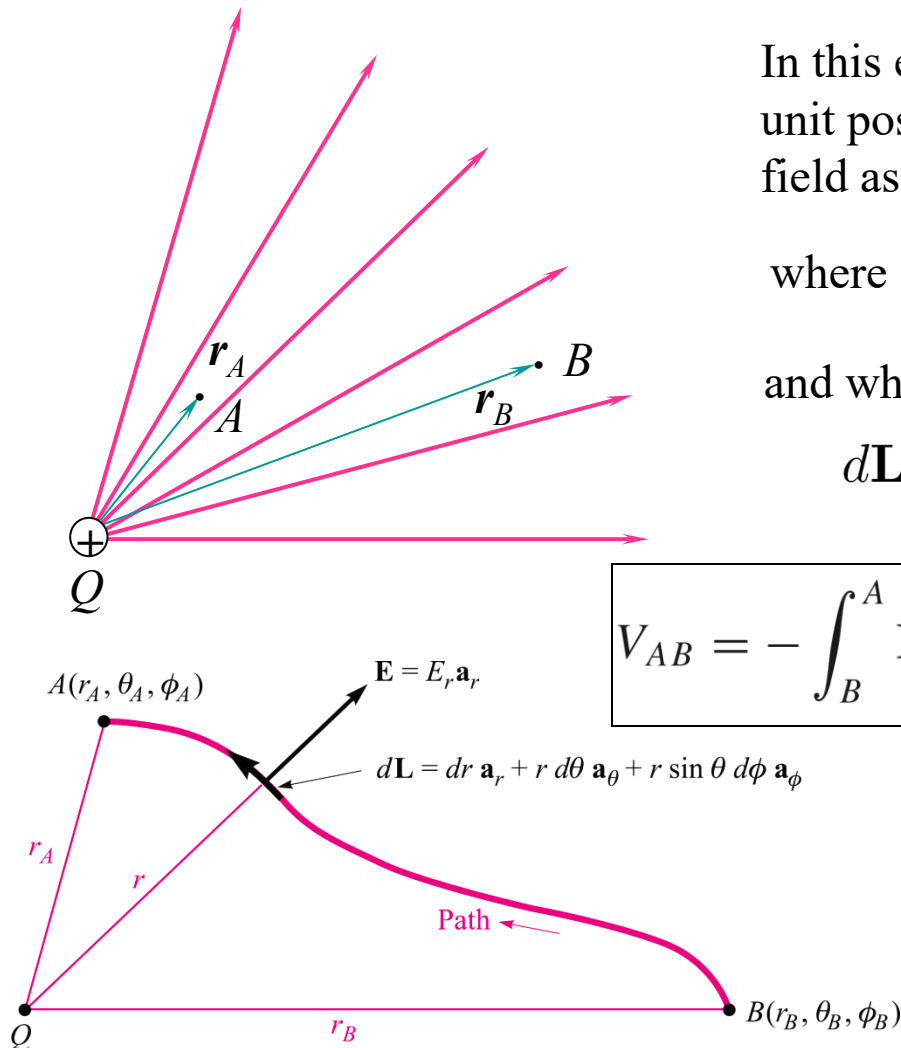
- Potential difference in a point charge field

In this exercise, we evaluate the work done in moving a unit positive charge from point B to point A , within the field associated with point charge Q

where $\mathbf{E} = E_r \mathbf{a}_r = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r$

and where in general:

$$d\mathbf{L} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$



$$V_{AB} = - \int_B^A \mathbf{E} \cdot d\mathbf{L} = - \int_{r_B}^{r_A} \frac{Q}{4\pi\epsilon_0 r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right)$$

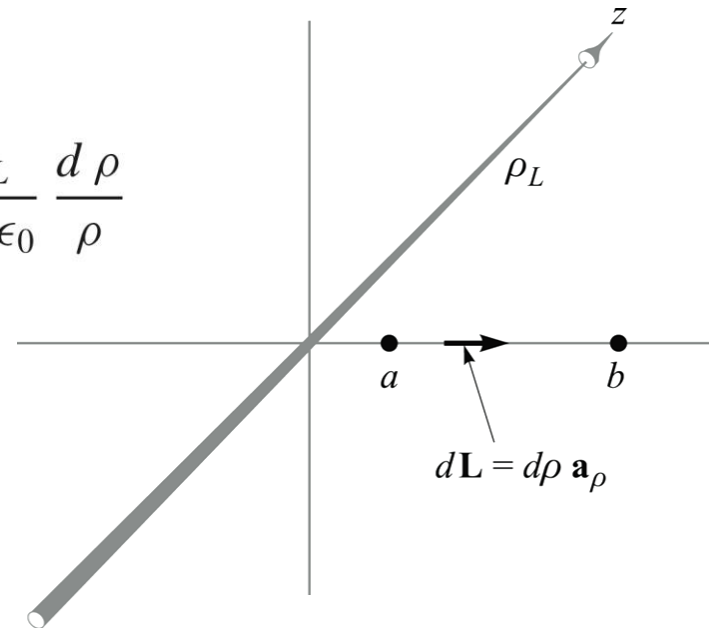
8: Energy and Potential 能量与电位



- Potential difference in a line charge field

$$W = -Q \int_{\text{init}}^{\text{final}} \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho \cdot d\rho \mathbf{a}_\rho = -Q \int_a^b \frac{\rho_L}{2\pi\epsilon_0} \frac{d\rho}{\rho}$$

$$W = -\frac{Q\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$



Thus, the potential difference between points at $\rho = a$ and $\rho = b$ is

$$V_{ab} = \frac{W}{Q} = \frac{\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$

8: Energy and Potential 能量与电位

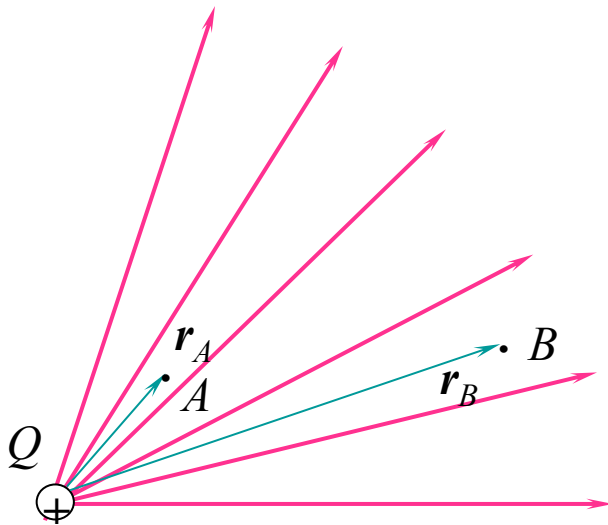


- Potential, or Absolute Potential 电位，绝对电位

It is often convenient to speak of the *potential*, or *absolute potential*, of a point, rather than the potential difference between two points, but this means only that we agree to measure every potential difference with respect to a specified reference point that we consider to have zero potential. 要求我们定义一个参考零电位点/面，如同重力场中的海平面（海拔高度=0）

Perhaps the most universal zero reference point in experimental or physical potential measurements is “ground,” 大地 by which we mean the potential of the surface region of the earth itself.

Another widely used reference “point” is a point which is infinitely far away (infinity). 无穷远处的点



$$V_{AB} = - \int_B^A \mathbf{E} \cdot d\mathbf{L} = - \int_{r_B}^{r_A} \frac{Q}{4\pi\epsilon_0 r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right)$$